4 Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Code

```
from sklearn.neural_network import MLPClassifier

X = [[0, 0], [0, 1], [1, 0], [1, 1]]
y = [0, 1, 1, 0]  # XOR output

model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='tanh',
    solver='lbfgs',
    max_iter=5000,
    random_state=1
)

model.fit(X, y)

print("Predictions:", model.predict(X))
print("Accuracy:", model.score(X, y))
```

Output:

